



SPECIFICATIONS OF SFD32KX8M64P4

Rev 010C

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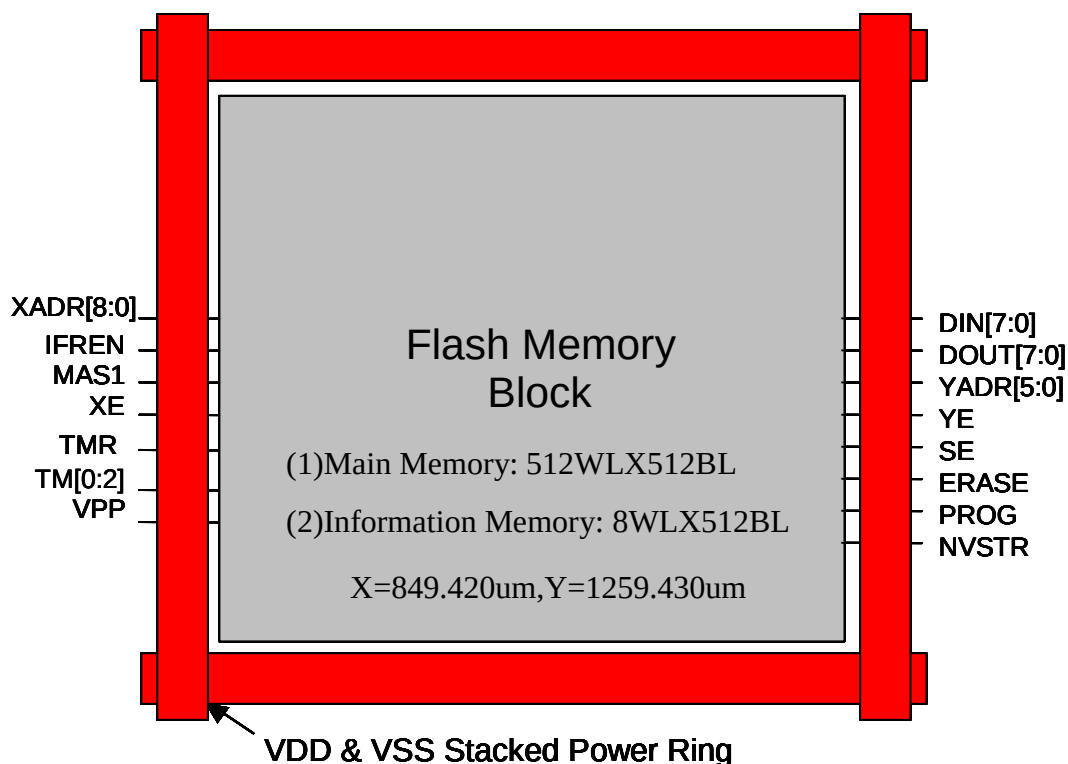
General Description

The SFD32KX8M64P4 is a CMOS page erase, byte (8-bit) program Embedded Flash Memory which is partitioned into two memory blocks. One is the main memory block, the other is the information block. The main memory block is organized as 32,768-word by 8-bit. The information block is organized as 512-word by 8-bit. The information block can be used for the storage of device information. The page erase operation erases all bytes within a page. A page is composed of four adjacent rows for main memory block and information block. The split gate cell design and thick oxide tunneling injector attain better reliability and manufacturability compared with alternative approaches. The SFD32KX8M64P4 erases and programs with a 1.8-volt only power supply.

Features

- Single 1.8-volt power supply
- Endurance: 1000 cycles (min) with 20ms erase and 20us program time at 1.8Vcc
- Memory Organization 32Kx8 + 512x8
- Greater than 100 years Data Retention under Room Temperature
- Page Erase Capability: 256 bytes per page
- Fast Page Erase/Word Program Operation
- Access Time : 25ns(max)
- Word Program Time: 20us(min)
- Current
 - Operating : 13mA(max)
 - Standby : 80uA(max)
 - Page Erase Time: 20 ms(min)
 - Mass Erase Time: 20 ms(min)

Block Diagram



Connector Description

User Mode¹

Connector name	Direction	Description
XADR[8:0]	I	X address input, access rows, XADR[1:0] select one row within a page of main memory block or information block.
YADR[5:0]	I	Y address input, access a byte within a row.
DIN[7:0]	I	Data input bus.
DOUT[7:0]	O	Data output bus.
XE	I	X address enable, all rows are disabled when XE=0.
YE	I	Y address enable, YMUX is disabled when YE=0.
SE	I	Sense amplifier enable.
IFREN ³	I	Information block enable
ERASE	I	Defines erase cycle.
MAS1	I	Defines mass erase cycle, erase whole block.
PROG	I	Defines program cycle.
NVSTR	I	Defines non-volatile store cycle.
VDD ²	I	Power supply
VSS ²	I	Ground

Notes:

1. Need to multiplex all control, address and data signals to package pins to facilitate direct memory test.
2. If routing power width is as wide as 30um, connecting only one set of VDD/VSS is allowable. If not, the total width of power line connecting to two sets or three sets of VDD/VSS should not be less than 30um .
3. When access the information block, the XADR[1:0] select one row within a page and XADR[8:3] are don't care.

Test Mode¹

Connector name	Direction	Description
TMR ²	I	Register reset for test mode. In user mode, it should be connected to VDD.
VPP	I/O	High voltage pin used in test mode and operating in 0~13V. A high voltage pad with a I/O path for VPP pin will be provided by TSMC.
TM[1:0]	I/O	Analog pins used in test mode and operating in 0~VDD. These analog pins are high impedance in user mode.
TM[2] ³	I/O	Analog pins used in test mode and operating in 0~3V. Since the operation voltage is higher than VDD, the high voltage pad is recommended to use for this pin.

Notes:

1. Need to multiplex the test mode related signals to package pins for memory testing.
2. TMR is a digital pin and operating in 0~VDD.
3. TM[2] is used for debugging and will not be used in testing flow and user mode.

Recommended DC Operating Conditions

(T_J = 0 °C to 125 °C)

Parameter	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage	V _{DD}	1.62	1.8	1.98	V
Ground	V _{SS}	0	0	0	V

DC Electrical Characteristics (T_J = 0 °C to 125 °C, V_{DD} = 1.62 V to 1.98 V, V_{SS} = 0V)

Parameter	Symbol	SFD32KX8M64P4		Unit	Condition
		Min.	Max.		
Operating Current Read Program Erase Mass erase	I _{DD1}	- - - -	13 5 7 5	mA	Min.Cycle,Duty= 100% V _{IN} = V _{DD} or V _{SS} I _{I/O} = 0mA
Static Read Current	I _{DD2}	-	5	mA	XE=YE=SE =V _{DD} I _{I/O} = 0mA
Standby Power Supply Current	I _{SB}	-	80	uA	XE, YE, SE, PROG, ERASE, and MAS1 are all V _{SS} . Other V _{IN} = V _{DD} or V _{SS}

Truth Table

Mode	XE	YE	SE	PROG	ERASE	MAS1	NVSTR
Standby	L	L	L	L	L	L	L
Read	H	H	H	L	L	L	L
Byte Program	H	H	L	H	L	L	H
Page Erase	H	L	L	L	H	L	H
Mass Erase	H	L	L	L	H	H	H

IFREN Truth Table

Mode	IFREN=1	IFREN=0
Read	Read information block	Read main memory block
Byte program	Program information block	Program main memory block
Page erase	Erase information block	Erase main memory block
Mass erase	Erase both block	Erase main memory block

Test Mode Decoder Truth Table

Mode	CODE				
	ERASE	YE	XE	IFREN	MAS1
Test mode 1	L	H	L	L	H
Test mode 2	L	H	L	H	L
Test mode 3	H	H	H	L	L
Test mode 4	H	H	H	L	H
Test mode 5	H	H	H	H	L

Capacitance

Parameter	Symbol	Min.	Max.	Unit
Address & Control Input pin Capacitance	C _{IN}	-	0.3	pF
Data Input pin Capacitance	C _{DIN}	-	0.3	pF
Data Output pin Capacitance	C _{DOUT}	-	0.1	pF

Timing Parameters ¹(T_J = 0 °C to 125 °C, V_{DD} = 1.62 V to 1.98 V, V_{SS} = 0V, C_{LOAD}=1pF)

User mode

Parameter	Symbol	SFD32KX8M64P4		Unit
		Min.	Max.	
X address access time	T _{xa}	-	25	ns
Y address access time	T _{ya}	-	25	ns
PROG/ERASE to NVSTR set up time	T _{nvs}	5	-	us
NVSTR hold time	T _{nvh}	5	-	us
NVSTR hold time(mass erase)	T _{nvh1}	100	-	us
NVSTR to program set up time	T _{pgs}	10	-	us
program hold time	T _{pgh}	20	-	ns
program time	T _{prog}	20	40	us
address/data set up time	T _{ads}	20	-	ns
address/data hold time	T _{adh}	20	-	ns
recovery time	T _{rcv}	1	-	us
cumulative program HV period ²	T _{hv}	-	8	ms
Erase time	T _{erase}	20	40	ms
Mass erase time	T _{me}	20	40	ms

Notes:

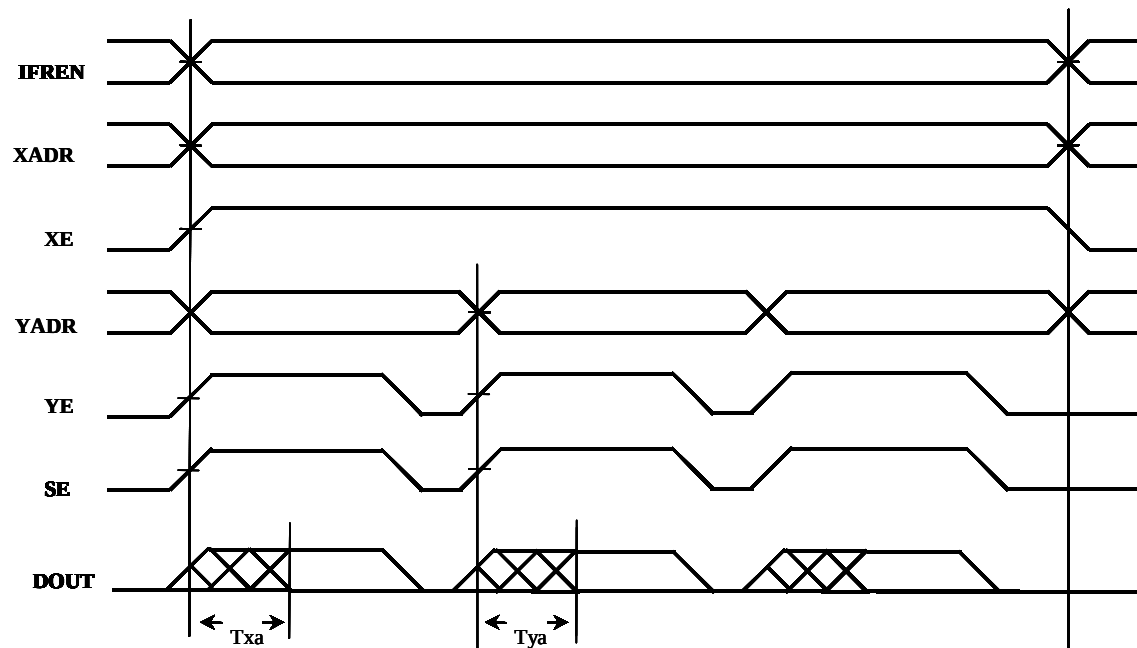
1. All timing numbers are pre-simulation and subject to change before silicon verification.
2. T_{hv} is the cumulative high voltage programming time to the same row before next erase and the same address can't be programmed more than twice before next erase.
3. All signals that align together in the timing diagrams should be derived from the same clock edge. Set up and hold times are not critical if they are within 10ns.
4. Reading operation is combinatorial. Sequences of timing edges are not important. The latest valid signal determines the access time.
5. All timing measurements are from the 50% of the input to 50% of the output.
6. All input waveforms have rising time(tr) and falling time(tf) of 1ns.
7. For capacitive loads greater than 1pF, access time will increase by 1ns per pF of additional loading.

Test mode

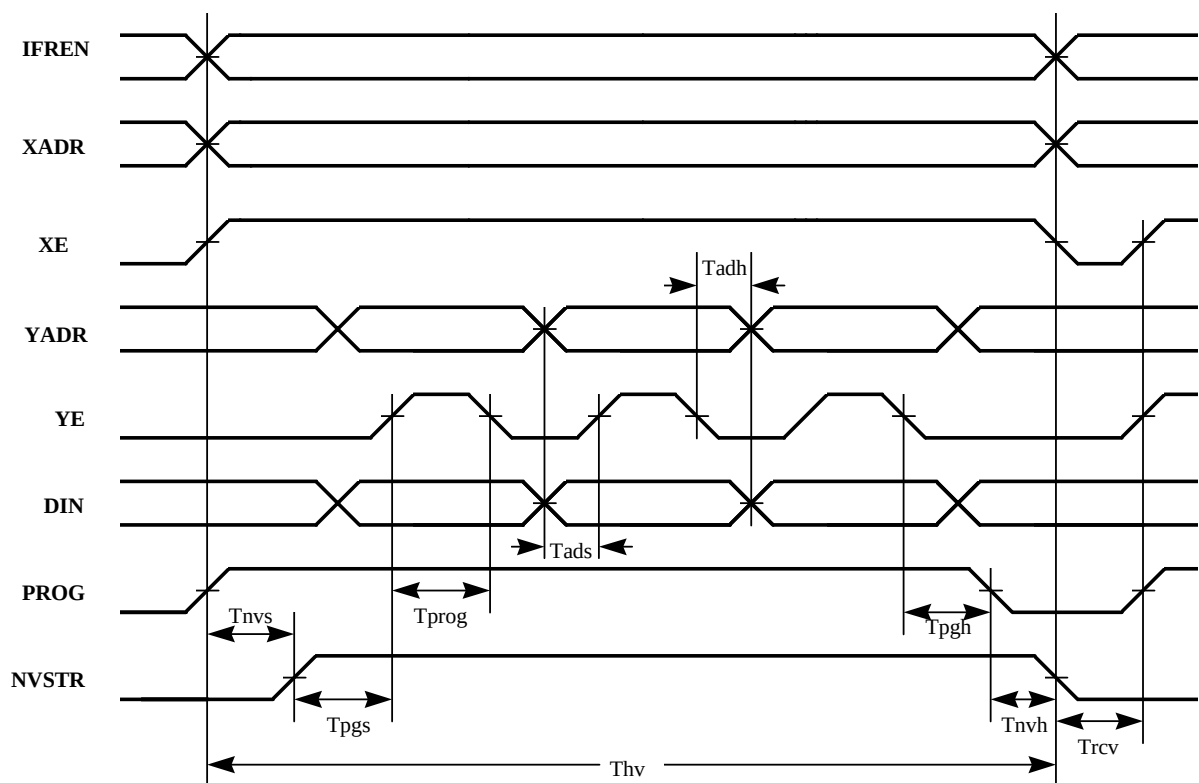
Parameter	Symbol	Value		Unit
		Min	Max	
TMR pulse width	T _{tmr}	20	-	ns
TMR to internal TMSE set up time	Trses	10	-	ns
Internal TMSE to data set up time	Tseds	10	-	ns
Data set up time for latch	Tlds	10	-	ns
Set latch pulse width	Tlpw	20	-	ns
Data hold time for latch	Tldh	10	-	ns
Data to internal TMSE hold time	Tdseh	10	-	ns

Timing Waveforms

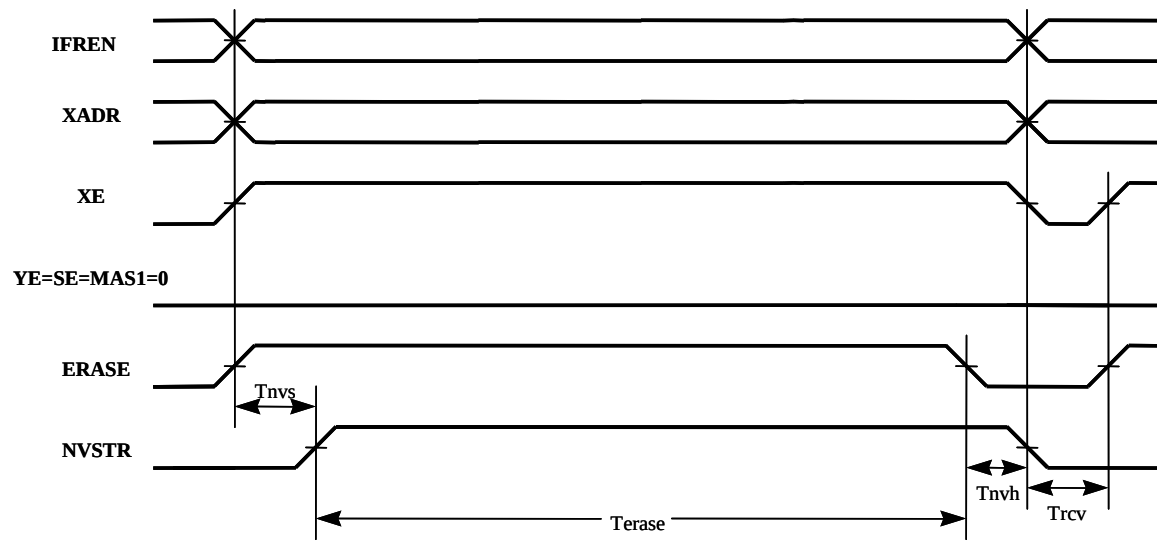
Read Cycle



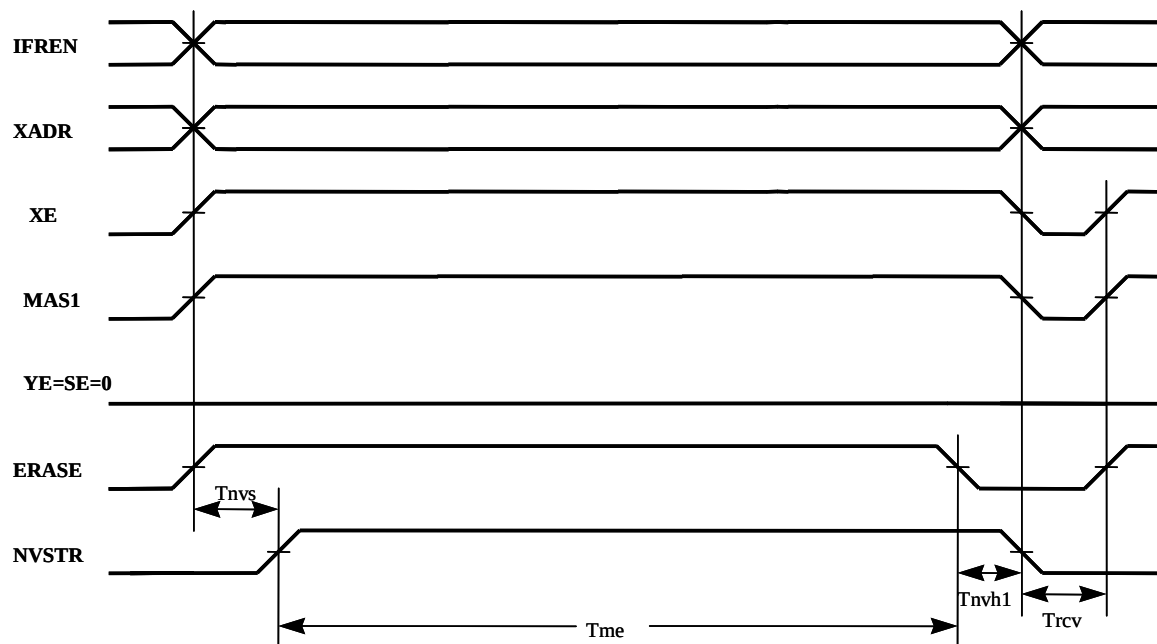
Program Cycle



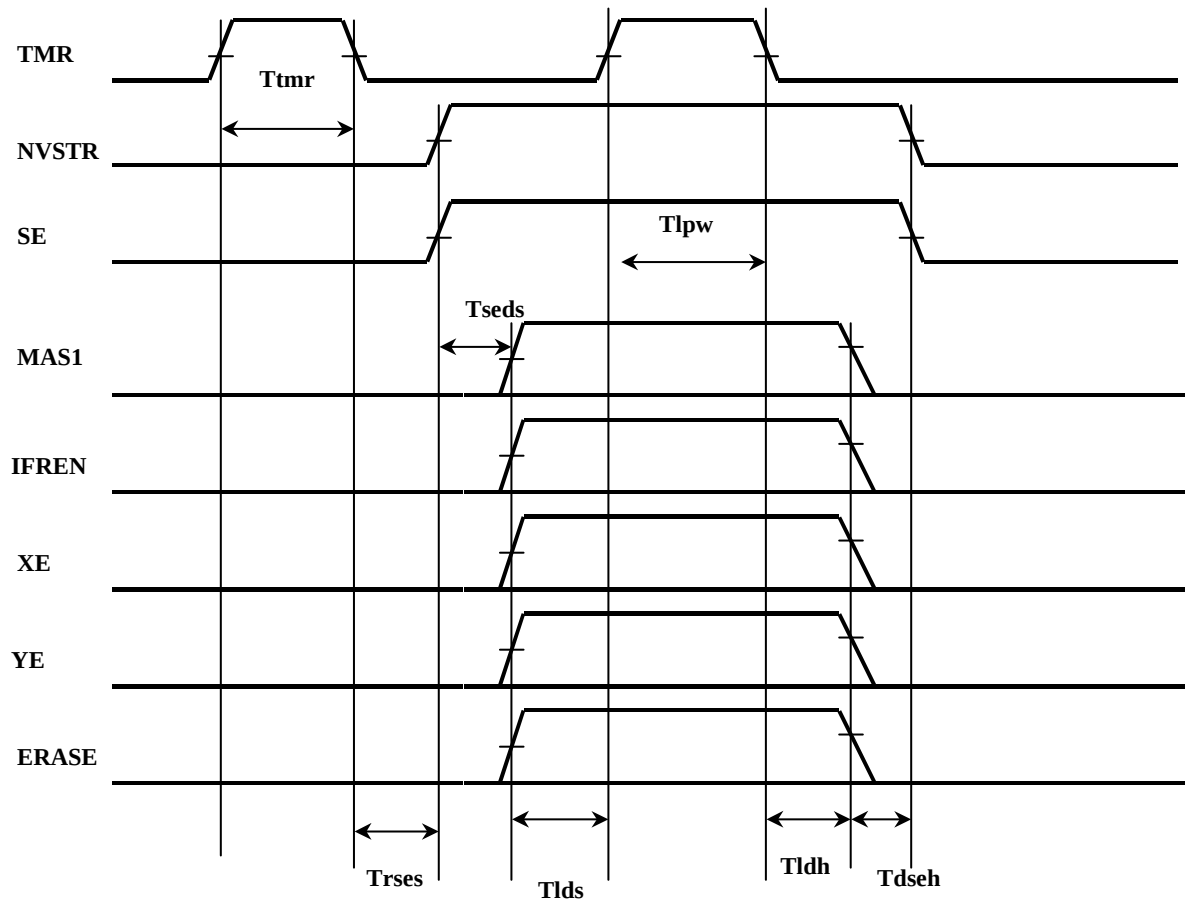
Erase Cycle



Mass Erase Cycle



Test mode



Update history:

Date	Rev	What	Who
12/10/2003	010A	New created.	SSLiu
04/20/2004	010B	Change endurance spec, mass erase time, access time, read operation current and test mode pin number TM[2].	SSLiu
05/21/2003	010C	1. Add qualification condition for 1K endurance 2. Update Terase, Tme	HHCHENA